

STA141B-HW5

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Assignment 5:

Computes the total number of questions from the front page

Total Questions: 219494

- [1] “<https://stats.stackexchange.com/questions/676169/how-can-i-improve-my-classifier>”
- [2] “<https://stats.stackexchange.com/questions/676168/busco-un-art%c3%adculo-cient%c3%adfico-de-agronomia-que-tenga-un-error-estadistico-f%c3%a1cil-d>” [3] “<https://stats.stackexchange.com/questions/676165/can-a-semi-parametric-survival-model-produce-survival-and-hazard-estimates-past>”
- [4] “<https://stats.stackexchange.com/questions/676163/what-is-the-optimal-variance-of-the-forecasts-according-to-yates>”
- [5] “<https://stats.stackexchange.com/questions/676162/model-with-one-latent-construct-and-multiple-observed-items-regression-best-app>”
- [6] “<https://stats.stackexchange.com/questions/676158/causalimpact-on-a-time-series-with-no-control-variable>”
- [7] “<https://stats.stackexchange.com/questions/676157/error-on-parameters-for-nonlinear-least-squares>”
- [8] “<https://stats.stackexchange.com/questions/676156/zero-degrees-of-freedom>”
- [9] “<https://stats.stackexchange.com/questions/676154/how-should-i-handle-lower-and-upper-assay-limits-when-calculating-a-correlation>”
- [10] “<https://stats.stackexchange.com/questions/676151/echo-state-networks-as-universal-approximators-bridging-theoretical-fading-memo>”
- [11] “<https://stats.stackexchange.com/questions/676150/cox-regression-linearity-for-time-dependent-covariate>”
- [12] “<https://stats.stackexchange.com/questions/676149/how-can-i-get-better-face-details-in-comfyui>”
- [13] “<https://stats.stackexchange.com/questions/676145/how-to-reduce-dimensionality-of-biological-ratios-when-pca-is-not-appropriate>”
- [14] “<https://stats.stackexchange.com/questions/676144/bayesian-statistics-proof-that-hat-thetay-arg-max-theta-f-theta>”
- [15] “<https://stats.stackexchange.com/questions/676143/can-you-statistically-compare-findings-across-two-separately-preregistered-exper>”

[1] 15 17 [1] 20 8 [1] 39 12 [1] 40 2

Table 1: Questions Table

	title	text	user_name	q_id	badges	reputation	date_posted	views	votes	question_id	comments_total
datetime...1	How can I improve my classifie	I have been researching using well known	Yash Banerjee	512569	OG / OS / OB	1.0	2026-06-04 21:52:20Z	25	0	676169	3
datetime...2	Busco un artAculo cientÁfico	In edge cases or for very opaque questi	ByPhotography	512565	OG / OS / OB	1.0	2026-06-04 19:27:31Z	18	-2	676168	1
datetime...3	Can a semi-parametric survival	I have a question about knowing if certa	adamkostanov	512554	OG / OS / 1B	13.0	2026-06-04 15:39:33Z	46	1	676165	0
datetime...4	What is the optimal variance o	Frank Yates introduced in [1] a covarian	Firebug	60613	1G / 1S / 1B	20.9	2026-06-04 15:37:44Z	35	2	676163	0
datetime...5	Model with one latent construc	I am exploring how product characteristi	user161652	161652	OG / 1S / 1B	13.0	2026-06-04 10:09:42Z	12	0	676162	1
datetime...6	CausalImpact on a Time series	Today I have been experimenting with the	Faz_Fiz	512523	OG / OS / 1B	11.0	2026-06-03 23:32:40Z	37	1	676158	0

Table 2: Tags Table

question_id	tag
676169	machine-learning
676169	classification
676168	regression
676165	survival
676163	forecasting
676163	mse
676163	scoring-rules
676162	regression
676162	multiple-regression
676162	predictive-models

Table 3: Comments Table

question_id	text	user_id	user_name	reputation	date_string	score
href...1676169	Welcome. With 4000 samples you might not be able to do better than what you have. Brain data are pre	97844	Rick Hass	8,143 reputation	2026-06-04 22:13:37Z, License: CC BY-SA 4.0	0
href1...376169	1) Accuracy as a measure of classifier performance is a very poor metric (it is affected by prevalen	380096	ginestet	17,357 reputation	2026-06-04 23:50:58Z, License: CC BY-SA 4.0	0
href2...376169	Sounds like you are overfitting to the validation set (yes, that is a thing). Take a look at this th	1352	Stephan Ko-lassa	141,649 reputation	2026-06-05 06:02:05Z, License: CC BY-SA 4.0	0
href...4676168	Welcome to CV. No matter what language this question is posed in, it's not on topic here, but you m	919	whuber	347,572 reputation	2026-06-04 19:55:46Z, License: CC BY-SA 4.0	0
href...5676163	+1 are going to send that preprint in for review? Looks nice from my cursory peak at it!	97844	Rick Hass	8,143 reputation	2026-06-04 15:52:31Z, License: CC BY-SA 4.0	3
href...6676162	Welcome to CV. What's your product? What are your characteristics? How are they measured? What's you	686	Peter Flom	145,620 reputation	2026-06-04 10:49:18Z, License: CC BY-SA 4.0	0

Table 4: Answers Table

text_preview	user_name	answers_count	reputation	date_posted	date_modified	is_accepted	answers_count	question_id	comments_count
Importantly, the semiparametric (Cox proportional hazards) model does not produce survival or hazard	Adam60131G	68.5	2026-06-04 15:37:44Z	2026-06-04 16:19:48Z	TRUE	4	676166	676165	0
An alternative arrangement of the terms shows that									

/begin{align*} /mathrm{BS} \} \&= /underbrace{\{ (|Firebug |60613 |1G / 1S / 1B | 20.9|2026-06-04 15:37:44Z |2026-06-04 18:53:26Z |FALSE | 2|676164 |676163 | 1| |datetime...17 |Try extending the characteristic function accordingly The approach used by Siegel (1979) is equivalent |Ben |173082 |1G / 1S / 1B | 146.0|2026-06-04 02:23:37Z |2026-06-04 22:42:05Z |FALSE | 5|676159 |676156 | 0| |datetime2 |Ben obviously gave good details based on Siegel's original paper, but I was intrigued by this magical |Rick Hass |97844 |1G / 1S / 1B | 8143.0|2026-06-04 18:30:42Z |2026-06-05 00:45:23Z |FALSE | 4|676167 |676156 | 0| |datetime...28 |Welcome to Cross Validated! The problem, as you recognize, is that the hazard for an individual cha |EdM |28500 |1G / 1S / 1B | 119.0|2026-06-03 14:50:45Z |2026-06-03 14:50:45Z |FALSE | 3|676153 |676150 | 2| |datetime...34 |The paper by Atchley et al., here cited fully, William R. Atchley, Charles T. Gaskins, Dwane Anders |Nick Cox |22047 |1G / 1S / 1B | 63.8|2026-06-02 21:35:27Z |2026-06-04 09:35:21Z |FALSE | 12|676147 |676145 | 3|

Table 5: Combined Table

file	title	user_name	date_posted	date_modified	is_accepted	answers_count	question_id	comments_count
How can I improve my classifier?	Adam60131G	68.5	2026-06-04 15:37:44Z	2026-06-04 16:19:48Z	TRUE	4	676166	676165
Busco un artículo científico de agronomía que te								
Can a semi-parametric survival model produce survival								
What is the optimal variance of the forecasts according to								
Model with one latent construct and multiple observed								
CausalImpact on a Time series with no control variables								
Error on parameters for nonlinear least squares								
Zero degrees of freedom								

```
## FILE [1]: Q-ID: 676169 | Title: 'How can I improve my classifier'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [2]: Q-ID: 676168 | Title: 'Busco un artículo científico de agronomía que te'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [3]: Q-ID: 676165 | Title: 'Can a semi-parametric survival model produce survival'
## --> Question has no code blocks.
## --> Found 1 answer:
## - Answer [676166]: has no code blocks.
## FILE [4]: Q-ID: 676163 | Title: 'What is the optimal variance of the forecasts according to'
## --> Question has no code blocks.
## --> Found 1 answer:
## - Answer [676164]: has no code blocks.
## FILE [5]: Q-ID: 676162 | Title: 'Model with one latent construct and multiple observed'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [6]: Q-ID: 676158 | Title: 'CausalImpact on a Time series with no control variables'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [7]: Q-ID: 676157 | Title: 'Error on parameters for nonlinear least squares'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [8]: Q-ID: 676156 | Title: 'Zero degrees of freedom'
## --> Question has no code blocks.
```

```

## --> Found 2 answer:
##     - Answer [676159]: has no code blocks.
##     - Answer [676167]: has no code blocks.
## FILE [9]: Q-ID: 676154 | Title: 'How should I handle lower and upper assay limits w'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [10]: Q-ID: 676151 | Title: 'Echo State Networks as universal approximators: br'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [11]: Q-ID: 676150 | Title: 'Cox regression linearity for time-dependent covari'
## --> Question has no code blocks.
## --> Found 1 answer:
##     - Answer [676153]: has no code blocks.
## FILE [12]: Q-ID: 676149 | Title: 'How can I get better face details in ComfyUI? [clo'
## --> Question has no code blocks.
## --> No answers posted.
## FILE [13]: Q-ID: 676145 | Title: 'How to reduce dimensionality of biological ratios '
## --> Question has no code blocks.
## --> Found 1 answer:
##     - Answer [676147]: has no code blocks.
## FILE [14]: Q-ID: 676144 | Title: 'Bayesian Statistics - Proof that  $\hat{\theta}(y)$  '
## --> Question has no code blocks.
## --> No answers posted.
## FILE [15]: Q-ID: 676143 | Title: 'Can you statistically compare findings across two '
## --> Question has no code blocks.
## --> No answers posted.

```

Describe how to arrange the results in a form that we can analyze it and address questions similar to assignment 3.

We can use the question id as a primary key across questions, user id as foreign key. Analyze views, votes, and date.

Across tags, question id as foreign key, linking it to the questions, its parent thread.

Use the answer id as a primary key across answers, question id as foreign key, linking it to the questions, its parent thread. Analyze acceptance, votes, and date.

Across comments, question id as foreign key, linking it to the questions, its main parent thread. If there's an answer id to it, its linked to the answer, if not, it links directly to the question. Analyze score and date.

To analyze and address questions similar to assignment 3, we can join tables and mimic that of relational databases. Some relations we can study is that between question post time and votes/views, post time and average speed of answer/comment responses, number of tags and view count, Answer acceptance and vote count. We can also analyze relation between user information and quality of their output, whether users with higher reputation and better badges post questions with more votes and views, answers with more acceptance and votes, and comments with higher scores.

For the question count, I initially wanted to find sum by using page count and question per page number, but this is inefficient as we'd deal with the constantly updated and changing of numbers, especially on the last page. In the end, I use the text on the front page and scrape it out as a integer to answer said question.

I initially ran into the error having `parse_question_file` return only null because of the `absolute_links`, pasting Markdown formatted hyperlink string as a literal prefix instead of base URL.

I couldn't get the updated and last modified time correctly and had to revisit.

After running into empty string returns, I chose to set default `NA_character_` for `q_id` and `a_id` to avoid said issue.

I originally used a count system for badge status but this was too weak, its been later updated to be a counter between the three tiers, showing a count for all three as a gold badge owner, also good for us to do validations on.

I was using downloaded HTML pages stored in my dictionary, I later added `force_refresh` to update with live web data.

I used Gemini, chat link: <https://gemini.google.com/app/4974e4a114a85d5f>

With Gemini, I gave verbal descriptions of what I want to achieve and ask it for code samples and suggestions, I check the code output by trying it and visualizing output segment by segment, I also ask it to explain the output code line by line so I'm not just copy pasting, but learning the code and understanding each line and function's purpose.

There were many redundancies in code that was given as feedback, I also made changes to the code to better fit the structure of the data, edited and reformatted for personal consistency and preference, removing identical library blocks and data blocks.

Some validation methods include manually comparing random specific questions to the website information, having both visible and invisible comment counts to validate total comment count, having the combined table where we can visualize comment and answer matching up with questions by id, also to see if there are any unreasonable NAs.

Validation below checks if questions having higher votes is correlated with having more views.

Correlation between Question Votes and Views: 0.715 -> Pass: Positive alignment

Validation below checks if users answering more questions correlate with scoring higher on average.

user_name	answers_provided	avg_answer_score
1 Nick Cox	2	6.0
2 Ben	2	2.5
3 AdamO	2	2.0
4 Rick Hass	2	2.0
5 EdM	2	1.5
6 Firebug	2	1.0

I worked on this assignment on and off throughout the weeks, around 2 hours everyday since the Tuesday before the deadline.

Appendix: R code

```
knitr::opts_chunk$set(echo = TRUE)
library(XML)
library(dplyr)
library(stringr)
library(knitr)
library(kableExtra)

escape_latex_strings <- function(text_vector) {
  if (is.character(text_vector)) {
    text_vector <- gsub("\\\\", "/", text_vector)      # backslashes first
    text_vector <- gsub("_", "\\_", text_vector)
    text_vector <- gsub("%", "\\%", text_vector)
    text_vector <- gsub("\\$", "\\$", text_vector)
    text_vector <- gsub("\\^", "\\^{}", text_vector)
    text_vector <- gsub("&", "\\&", text_vector)
    text_vector <- gsub("#", "\\#", text_vector)
    text_vector <- gsub("\\{", "\\{", text_vector)
    text_vector <- gsub("\\}", "\\}", text_vector)
    text_vector <- gsub("~", "\\textasciitilde{}", text_vector)
    text_vector <- gsub("<", "\\textless{}", text_vector)
    text_vector <- gsub(">", "\\textgreater{}", text_vector)
  }
  return(text_vector)
}

sanitize_df_for_latex <- function(df) {
  if (nrow(df) == 0) return(df)
  # Safely sanitize column headers
  colnames(df) <- gsub("_", "\\_", colnames(df))
  # Safely sanitize cell contents
  df <- df %>% mutate(across(where(is.character), escape_latex_strings))
  return(df)
}

#manual change force refresh or not
force_refresh <- TRUE

cache_dir <- "stats_cache_html"
if (!dir.exists(cache_dir)) dir.create(cache_dir)

first_page_file <- file.path(cache_dir, "list_page_1.html")
if (!file.exists(first_page_file)) {
  live_url <- "https://stats.stackexchange.com/questions?tab=newest&page=1"
  download.file(live_url, destfile = first_page_file, quiet = TRUE)
}

initial_doc <- htmlParse(first_page_file, isURL = FALSE)

total_questions_raw <- xpathSApply(initial_doc, "//div[contains(@class, 'd-flex') and contains(@class,

#get total number on front page
clean_number_string <- gsub("[^0-9]", "", total_questions_raw[1])
global_total_questions <- as.integer(clean_number_string)
cat("### Total Questions:", global_total_questions, "\n")
```

```

all_question_urls <- character()
current_page <- 1

#manual change number of questions you want to use
N <- 15

#find number of questions per page and default use that
page_size_raw <- xpathSApply(initial_doc, "//div[contains(@class, 'page-sizer')]/a[contains(@class, 'is')]")
detected_page_size <- if(length(page_size_raw) > 0) as.numeric(trimws(page_size_raw[1])) else 15

max_questions <- if(!is.null(N)) N else detected_page_size

#loop through
while(length(all_question_urls) < max_questions) {

  local_page_file <- file.path(cache_dir, paste0("list_page_", current_page, ".html"))

  if (file.exists(local_page_file) && !force_refresh) {
    doc <- htmlParse(local_page_file, isURL = FALSE)
  } else {
    #get live
    live_url <- paste0("https://stats.stackexchange.com/questions?tab=newest&page=", current_page)

    Sys.sleep(runif(1, 1.5, 3.0))

    download.file(live_url, destfile = local_page_file, quiet = TRUE)
    doc <- htmlParse(local_page_file, isURL = FALSE)
  }

  page_links <- xpathSApply(doc, "//a[@class='s-link' and contains(@href, '/questions/')]/@href")
  if (length(page_links) == 0) break

  absolute_links <- paste0("https://stats.stackexchange.com", page_links)
  all_question_urls <- c(all_question_urls, absolute_links)
  current_page <- current_page + 1
}

all_question_urls <- head(all_question_urls, max_questions)
all_question_urls
question_dir <- "question_html_cache"
if (!dir.exists(question_dir)) dir.create(question_dir)

local_file_paths <- character(length(all_question_urls))

for (i in seq_along(all_question_urls)) {
  url <- all_question_urls[i]
  #replace special characters in the URL to underscore
  safe_name <- gsub("[^[:alnum:]]", "_", url)
  local_file <- file.path(question_dir, paste0(safe_name, ".html"))
  local_file_paths[i] <- local_file

  if (!file.exists(local_file) || force_refresh) {
    Sys.sleep(runif(1, 1.5, 3.0))
  }
}

```

```

    tryCatch({
      download.file(url, destfile = local_file, quiet = TRUE)
    }, error = function(e) {
      message("Error downloading URL: ", url)
    })
  }
}

safe_value <- function(nodes, xpath, is_numeric = FALSE) {
  #check if xpath dealing with HTML attribute
  if (grepl("/@[a-zA-Z]", xpath)) {
    #return raw
    val <- xpathSApply(nodes, xpath)
  } else {
    #return plain text between HTML tags
    val <- xpathSApply(nodes, xpath, xmlValue)
  }
  if (length(val) == 0 || is.null(val)) return(ifelse(is_numeric, 0, NA_character_))
  #keep numerical, convert, isolate first
  if (is_numeric) return(as.numeric(gsub("[^0-9.-]", "", val[1])))
  return(trimws(val[1]))
}

parse_question_file <- function(file_path) {
  doc <- htmlParse(file_path, isURL = FALSE)

  #question
  q_node <- getNodeSet(doc, "//div[@id='question']")
  if (length(q_node) == 0) {
    free(doc)
    return(NULL)
  }
  q_node <- q_node[[1]]
  q_id <- xmlGetAttr(q_node, "data-questionid", default = NA_character_)

  title_raw <- xpathSApply(doc, "//a[@class='question-hyperlink']/text()", xmlValue)
  title <- if(length(title_raw) > 0) title_raw[1] else NA_character_

  q_text_nodes <- getNodeSet(q_node, ".//div[contains(@class, 'js-post-body')]")
  q_text <- if(length(q_text_nodes) > 0) paste(xpathSApply(q_text_nodes[[1]], ".//p | .//li | .//code"))

  q_votes <- safe_value(doc, "//div[@id='question']//div[contains(@class, 'js-vote-count')]/@data-val")

  views_raw <- xpathSApply(doc, "//meta[@itemprop='viewCount']/@content")

  if (length(views_raw) == 0) {
    views_raw <- xpathSApply(doc, "//div[contains(@class, 'd-flex') and contains(@class, 'fw-wrap')]")
  }

  q_views <- if(length(views_raw) > 0) as.numeric(gsub("[^0-9]", "", views_raw[1])) else NA_real_
  q_date <- safe_value(doc, "//time[@itemprop='dateCreated']/@datetime")
  q_mod <- safe_value(doc, "//a[contains(@href, 'lastactivity')]/@title")
  q_user_node <- getNodeSet(q_node, ".//div[contains(@class, 'post-layout--right')]/div[contains(@class, 'user')]")
  q_user_name <- NA_character_; q_user_rep <- NA_real_; q_badges <- NA_character_
  if (length(q_user_node) > 0) {

```

```

asker_card <-
q_user_node[[length(q_user_node)]]
q_user_name <- safe_value(asker_card, ".*div[@class='user-details']/a")
q_user_rep <- safe_value(asker_card, ".*span[@class='reputation-score']", is_numeric = TRUE)

badge_gold <- length(getNodeSet(asker_card, ".*span[contains(@class,'badge1')]"))
badge_silver <- length(getNodeSet(asker_card, ".*span[contains(@class,'badge2')]"))
badge_bronze <- length(getNodeSet(asker_card, ".*span[contains(@class,'badge3')]"))
q_badges <- paste0(badge_gold, "G / ", badge_silver, "S / ", badge_bronze, "B")
}
q_user_href <- safe_value(asker_card, ".*div[@class='user-details']/a/@href")
q_user_id <- NA_character_
if (length(q_user_node) > 0 && exists("asker_card")) {
  q_user_href <- safe_value(asker_card, ".*div[@class='user-details']/a/@href")
  q_user_id <- if(!is.na(q_user_href)) str_extract(q_user_href, "(?<=/users/)\d+") else NA_character_
}

#comment count
q_comm_visible <- length(getNodeSet(q_node, ".*ul[contains(@class, 'comments-list')]//li[contains(text(), 'show')]"))
hidden_text <- safe_value(q_node, ".*a[contains(@class, 'js-show-link')] or contains(text(), 'show')")
q_comm_hidden <- 0

if (!is.na(hidden_text) && hidden_text != "") {
  q_comm_hidden <- as.numeric(gsub("[^0-9]", "", hidden_text))
  if (is.na(q_comm_hidden)) q_comm_hidden <- 0
}
q_comm_total <- q_comm_visible + q_comm_hidden

#code check
code_nodes <- getNodeSet(q_node, ".*pre/code")
has_code <- length(code_nodes) > 0
q_code_text <- NA_character_; q_code_lang <- NA_character_
if (has_code) {
  q_code_text <- paste(sapply(code_nodes, xmlValue), collapse = "\n\n--- next block ---\n\n")
  raw_langs <- sapply(code_nodes, function(n) xmlGetAttr(n, "class", default = "unknown"))
  q_code_lang <- paste(raw_langs, collapse = ", ")
}

df_question <- data.frame(
  question_id = q_id,
  title = title,
  text = q_text,
  votes = q_votes,
  views = q_views,
  date_posted = q_date,
  last_modified = q_mod,
  user_name = q_user_name,
  q_id = q_user_id,
  reputation = q_user_rep,
  badges = q_badges,
  comments_total = q_comm_total,

```

```

      comments_hidden = q_comm_hidden,
      comments_visible = q_comm_visible,
      has_code = has_code,
      code_text = q_code_text,
      code_language = q_code_lang,
      stringsAsFactors = FALSE
    )

    #question tags
    tag_nodes <- xpathSApply(q_node, "//*[contains(@class, 'post-tag') or contains(@class, 's-tag')]/t
    df_tags <- if(length(tag_nodes) > 0) {
      data.frame(question_id = q_id,
                 tag = tag_nodes,
                 stringsAsFactors = FALSE)
    } else {
      data.frame(question_id = character(), owner_id = character(), tag = character(), stringsAsFactor
    }

    #answers
    ans_nodes <- getNodeSet(doc, "//*[id='answers']//div[contains(@class, 'answer')]")
    list_answers <- list()
    if (length(ans_nodes) > 0) {
      for (i in seq_along(ans_nodes)) {
        a_node <- ans_nodes[[i]]
        a_id <- xmlGetAttr(a_node, "data-answerid", default = NA_character_)
        a_text <- safe_value(a_node, "//*[contains(@class, 'js-post-body')]")
        a_votes <- safe_value(a_node, "//*[contains(@class, 'js-vote-count')]/@data-value", is_n
        is_accepted <- !is.null(xmlGetAttr(a_node, "class")) && grepl("accepted-answer", xmlGetAttr
        a_date <- safe_value(a_node, "//*[time[@itemprop='dateCreated']/@datetime")
        a_last_modified <- safe_value(a_node, "//*[a[contains(@href, 'revisions')]]//span[contains(@cl
        a_comm_count <- length(getNodeSet(a_node, "//*[ul[contains(@class, 'comments-list')]]//li[con
        if (is.na(a_last_modified) || a_last_modified == "") {
          a_last_modified <- safe_value(a_node, "//*[div[contains(@class, 'user-status-modification
        }
        a_user_name <- NA_character_;
        a_user_id <- NA_character_;
        a_user_rep <- NA_real_;
        a_user_badges <- NA_character_
        if (is.na(a_last_modified) ||
            a_last_modified == "") {
          a_last_modified <- a_date
        }

        a_user_node <- getNodeSet(a_node, "//*[div[contains(@class, 'user-info')]]")
        a_user_name <- NA_character_; a_user_rep <- NA_real_; a_user_badges <- NA_character_
        if (length(a_user_node) > 0) {
          ans_card <- a_user_node[[length(a_user_node)]]
          a_user_name <- safe_value(ans_card, "//*[div[@class='user-details']/a")
          a_user_href <- safe_value(ans_card, "//*[div[@class='user-details']/a/@href")
          a_user_id <- if(!is.na(a_user_href)) str_extract(a_user_href, "(?<=/users/)\d+") else NA
          a_user_rep <- safe_value(ans_card, "//*[span[@class='reputation-score']", is_numeric = TRUE
          a_badge_counts <- xpathSApply(ans_card, "//*[span[contains(@class, 'badge')]]//span[@class='b
        if (length(a_badge_counts) > 0) a_user_badges <- paste(a_badge_counts, collapse = "/")
      }
    }
  }
}

```

```

a_gold   <- length(getNodeSet(ans_card, ".*span[contains(@class,'badge1')]"))
a_silver <- length(getNodeSet(ans_card, ".*span[contains(@class,'badge2')]"))
a_bronze <- length(getNodeSet(ans_card, ".*span[contains(@class,'badge3')]"))
a_user_badges <- paste0(a_gold, "G / ", a_silver, "S / ", a_bronze, "B")
}

a_comm_count <- length(getNodeSet(a_node, ".*li[contains(@class, 'comment')]"))

list_answers[[i]] <- data.frame(
  answer_id = a_id,
  question_id = q_id,
  text = a_text,
  votes = a_votes,
  is_accepted = is_accepted,
  date_posted = a_date,
  last_modified = a_last_modified,
  user_name = a_user_name,
  ans_id = a_user_id,
  reputation = a_user_rep,
  badges = a_user_badges,
  comments_count = a_comm_count,
  stringsAsFactors = FALSE
)
}
}

df_answers <- if(length(list_answers) > 0) do.call(rbind, list_answers) else data.frame()

#comments
comm_nodes <- getNodeSet(doc, ".*li[contains(@class, 'comment')]")
list_comments <- list()
if (length(comm_nodes) > 0) {
  for (j in seq_along(comm_nodes)) {
    c_node <- comm_nodes[[j]]

    c_text <- safe_value(c_node, ".*span[contains(@class, 'comment-copy')]")
    c_user <- safe_value(c_node, ".*a[contains(@class, 'comment-user') or @itemprop='author']")

    c_score <- xmlGetAttr(c_node, "data-comment-score", default = "0")
    c_rep <- safe_value(c_node, ".*a[contains(@class, 'comment-user')]/@title")

    c_date <- safe_value(c_node, ".*span[contains(@class, 'comment-date')]/@title")
    if (is.na(c_date) || c_date == "") {
      c_date <- safe_value(c_node, ".*span[contains(@class, 'comment-date')]/a/@title")
    }

    c_user_href <- safe_value(c_node, ".*a[contains(@class, 'comment-user')]/@href")
    c_id <- if(!is.na(c_user_href)) {
      str_extract(c_user_href, "(?<=/users/)\d+")
    } else {
      NA_character_
    }

    parent_node <- xmlParent(xmlParent(c_node))
  }
}

```

```

is_q_comment <- !is.null(xmlGetAttr(parent_node, "id")) && grepl("question", xmlGetAttr(parent_node, "id"))
parent_id <- if(is_q_comment) q_id else safe_value(c_node, "ancestor::div[contains(@class, 'question')]")

list_comments[[j]] <- data.frame(
  question_id = q_id,
  type = ifelse(is_q_comment, "Question", "Answer"),
  text = c_text,
  user_name = c_user,
  user_id = c_id,
  reputation = c_rep,
  score = c_score,
  date_string = c_date,
  stringsAsFactors = FALSE
)
}
}
df_comments <- if(length(list_comments) > 0) do.call(rbind, list_comments) else data.frame()

free(doc)
return(list(question = df_question, tags = df_tags, answers = df_answers, comments = df_comments))
}
all_parsed_outputs <- lapply(local_file_paths, function(path) {
  if (file.exists(path)) return(parse_question_file(path))
  return(NULL)
})

all_parsed_outputs <- all_parsed_outputs[!sapply(all_parsed_outputs, is.null)]

final_questions_table <- bind_rows(lapply(all_parsed_outputs, function(x) x$question))
final_tags_table <- bind_rows(lapply(all_parsed_outputs, function(x) x$tags))
final_answers_table <- bind_rows(lapply(all_parsed_outputs, function(x) x$answers))
final_comments_table <- bind_rows(lapply(all_parsed_outputs, function(x) x$comments))

print(dim(final_questions_table))
print(dim(final_comments_table))
print(dim(final_answers_table))
print(dim(final_tags_table))
#limit text to 100 character long preview
q_table_preview <- final_questions_table[, c("title", "text", "user_name", "q_id", "badges", "reputation")]
transform(q_table_preview, text = substr(text, 1, 40), title = substr(title, 1, 30))

kable(head(sanitize_df_for_latex(q_table_preview)),
  format = "latex",
  booktabs = TRUE,
  caption = "Questions Table") %>%
  kable_styling(latex_options = c("striped", "scale_down", "hold_position"))
#most recent 10
kable(
  head(sanitize_df_for_latex(final_tags_table), 10),
  format = "latex",
  booktabs = TRUE,
  caption = "Tags Table"
) %>%

```

```

kable_styling(latex_options = c("striped", "hold_position"))
kable(head(sanitize_df_for_latex(final_comments_table[, c("question_id", "text", "user_id", "user_name")
  transform(text = substr(text, 1, 100))),
  caption = "Comments Table")
#clean NA entries, give text preview
answered_table_clean <- final_answers_table[!is.na(final_answers_table$answer_id), ]
answered_table_clean$text_preview <- substr(answered_table_clean$text, 1, 100)

kable(head(sanitize_df_for_latex(answered_table_clean[, c("text_preview", "user_name", "ans_id", "badges")
  caption = "Answers Table")
#combined table, group by question id
tags_collapsed <- final_tags_table %>%
  group_by(question_id) %>%
  summarise(tag_list = paste(tag, collapse = ", "), .groups = "drop")

#combine comments
comments_collapsed <- final_comments_table %>%
  group_by(question_id) %>%
  summarise(
    text_comment = paste(substr(text, 1, 15), collapse = " | "),
    score_comment = paste(score, collapse = ", "),
    .groups = "drop"
  )

#combine answers, remove duplicates
answers_prepped <- final_answers_table %>%
  filter(!is.na(answer_id) & answer_id != "") %>%
  distinct(question_id, answer_id, .keep_all = TRUE) %>%
  rename(
    text_answer = text,
    user_name_answer = user_name,
    reputation_answer = reputation,
    badges_answer = badges,
    votes_answer = votes,
    date_posted_answer = date_posted,
    last_modified_answer = last_modified
  )

#left join
flat_unified_table <- final_questions_table %>%
  left_join(tags_collapsed, by = "question_id") %>%
  left_join(comments_collapsed, by = "question_id") %>%
  left_join(answers_prepped, by = "question_id", relationship = "many-to-many")

available_cols <- colnames(flat_unified_table)

flat_unified_table <- flat_unified_table %>%
  mutate(
    text = if("text" %in% available_cols) substr(text, 1, 20) else "None",
    text_comment = if("text_comment" %in% available_cols) substr(text_comment, 1, 20) else "No Comment",
    title = if("title" %in% available_cols) substr(title, 1, 20) else "None",

#NAs

```

```

text_answer = if("text_answer" %in% available_cols) {
  ifelse(is.na(text_answer) | text_answer == "", "No Answers", substr(text_answer, 1, 20))
} else "No Answers",

  answer_id = if("answer_id" %in% available_cols) {
    ifelse(is.na(answer_id), "None", as.character(answer_id))
  } else "None"
)

target_columns <- c(
  "title", "text", "user_name", "q_id", "badges", "reputation", "date_posted", "last_modified", "tag_li
)

columns_to_show <- intersect(target_columns, available_cols)

kable(head(sanitize_df_for_latex(flat_unified_table[, columns_to_show]), 10),
  format = "latex",
  booktabs = TRUE,
  caption = "Combined Table") %>%
  kable_styling(latex_options = c("striped", "scale_down", "hold_position"))
#checking for code blocks
for (file_idx in seq_along(local_file_paths)) {
  path <- local_file_paths[file_idx]
  if (!file.exists(path)) next

  doc <- htmlParse(path, isURL = FALSE)

  q_node <- getNodeSet(doc, "//div[@id='question']")
  if (length(q_node) > 0) {
    q_node <- q_node[[1]]
    q_id <- xmlGetAttr(q_node, "data-questionid", default = "UNKNOWN")
    q_title <- safe_value(doc, "//a[@class='question-hyperlink']/text()")

    q_code_nodes <- getNodeSet(q_node, ".*pre/code")

    cat(sprintf("FILE [%d]: Q-ID: %s | Title: '%s'\n", file_idx, q_id, substr(q_title, 1, 50)))
    if (length(q_code_nodes) > 0) {
      q_lang <- xmlGetAttr(q_code_nodes[[1]], "class", default = "no-lang-tag")
      cat(sprintf(" --> [!] Question has code blocks: %d | First Lang: %s\n", length(q_code_nodes), q_lang))
      clean_q_snippet <- gsub("\\\\", "", xmlValue(q_code_nodes[[1]]))
      cat(sprintf("      Snippet: %s\n", gsub("\n", " ", substr(clean_q_snippet, 1, 80))))
    } else {
      cat(" --> Question has no code blocks.\n")
    }
  }

  ans_nodes <- getNodeSet(doc, "//div[@id='answers']//div[@data-answerid]")

  if (length(ans_nodes) > 0) {
    cat(sprintf(" --> Found %d answer:\n", length(ans_nodes)))

    for (a_idx in seq_along(ans_nodes)) {
      a_node <- ans_nodes[[a_idx]]

```

```

a_id <- xmlGetAttr(a_node, "data-answerid", default = "UNKNOWN")

a_code_nodes <- getNodeSet(a_node, ".*pre/code")

if (length(a_code_nodes) > 0) {
  a_lang <- xmlGetAttr(a_code_nodes[[1]], "class", default = "no-lang-tag")
  cat(sprintf("      - Answer [%s]: has code blocks | First Lang: %s\n", a_id, a_lang))
clean_a_snippet <- gsub("\\\\", "", xmlValue(a_code_nodes[[1]]))
cat(sprintf("      Snippet: %s\n", gsub("\n", " ", substr(clean_a_snippet, 1, 80))))
} else {
  cat(sprintf("      - Answer [%s]: has no code blocks.\n", a_id))
}
}
} else {
  cat(" --> No answers posted.\n")
}

free(doc)
}
if(nrow(final_questions_table) > 1) {
  correlation <- cor(final_questions_table$votes, final_questions_table$views, use = "complete.obs")
  cat(sprintf("Correlation between Question Votes and Views: %.3f\n", correlation))
  if(correlation > 0) {
    cat(" --> Pass: Positive alignment\n\n")
  } else {
    cat(" --> Fail: Flat or negative trend\n\n")
  }
}
}
user_quality_profile <- final_answers_table %>%
  filter(!is.na(user_name) & user_name != "") %>%
  group_by(user_name) %>%
  summarise(
    answers_provided = n(),
    avg_answer_score = mean(votes, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(avg_answer_score), desc(answers_provided)) %>%
  head(10)

print(as.data.frame(user_quality_profile))
cat("\n\nnewpage") #new page

```